

AI CONTRACT LIFECYCLE & COMPLIANCE REVIEW AGENT

USER GUIDE

Agentic AI Contract Analysis, RAG & Human-in-the-Loop Workflow

1. Purpose

This guide explains how to use the AI Contract Lifecycle & Compliance Review Agent. The system reviews contract terms against approved policy documents, retrieves relevant evidence, generates an evidence-grounded review using Gemini, routes deviations for human approval through n8n, and stores the review outcome in SQL Server.

2. What the System Does

- Processes contract and policy documents.
- Retrieves relevant sections using embeddings and FAISS.
- Uses Gemini to generate an evidence-grounded review.
- Produces a structured decision and recommended next step.
- Routes deviations to human review.
- Stores the AI finding, human decision and workflow outcome in SQL Server.

3. Technology Stack

Technology	Purpose
Python	Document processing, RAG pipeline and agent service
Jupyter Notebook	Development and validation
Sentence Transformers	Text embeddings
FAISS	Semantic retrieval
Google Gemini	LLM review generation
n8n	Workflow orchestration and human review
Streamlit	User interface
SQL Server / SSMS	Audit storage
GitHub	Version control and documentation

4. Main Workflow

Streamlit → n8n → Python/RAG → Gemini → Agent Decision → Human Review → Workflow Status → SQL Server Audit

1. Streamlit initiates the review.
2. n8n orchestrates the workflow.
3. Python/RAG retrieves relevant contract and policy evidence.
4. Gemini generates the grounded review.
5. The agent produces a structured decision.

6. Human reviewer approves, requests further review, or rejects.
7. n8n records the workflow outcome in SQL Server.

5. Prerequisites

- Python installed.
- Required project packages installed.
- Gemini API key available as GEMINI_API_KEY.
- SQL Server / SQL Server Express available.
- n8n installed and available locally.
- Project contract and policy documents available.

6. Start the Python Review Service

Open PowerShell in the project folder and set the Gemini key as an environment variable.

```
$env:GEMINI_API_KEY="YOUR_KEY"
```

Do not place the real key in source code, screenshots or GitHub.

```
python contract_review_runner.py
```

7. Start n8n

```
n8n.cmd
```

Open the local n8n editor. The configured workflow connects the review request to Python, routes the AI decision, opens human review when required, and writes the result to SQL Server.

8. Start Streamlit

```
streamlit run app.py
```

Streamlit provides the review interface. The current implementation initiates the demonstrated end-to-end workflow using the configured contract/policy knowledge sources.

9. Run a Contract Review

8. Open Streamlit.
9. Use the contract for the demonstrated review scenario.
10. Enter: "Does this contract comply with the standard payment policy?"
11. Click the review button.
12. Verify the n8n execution.
13. The Python/RAG service retrieves evidence and Gemini generates the review.

10. Demonstrated Scenario & Output

Item	Value
Contract payment term	90 calendar days
Policy standard	30 calendar days

Decision status	DEVIATION_FOUND
Next step	HUMAN_REVIEW_REQUIRED

The demonstrated scenario identifies a non-standard payment term and routes it to human review.

11. Human Review

When a deviation is identified, the n8n Wait step presents the human-review form.

- APPROVE_EXCEPTION
- REQUEST_FURTHER_REVIEW
- REJECT_EXCEPTION

The authorized human reviewer retains the final business/legal decision.

12. SQL Server Audit

After the human decision, n8n maps the result into Contract_Review_Audit. The record includes the review question, AI decision, recommended next step, reviewer, reviewer decision, workflow action/status, contract evidence, policy evidence and agent review.

```
SELECT TOP 5 *
FROM Contract_Review_Audit
ORDER BY Review_ID DESC;
```

13. End-to-End Demonstration

14. Start the Python service.
15. Start n8n.
16. Start Streamlit.
17. Initiate the review from Streamlit.
18. Verify DEVIATION_FOUND and HUMAN_REVIEW_REQUIRED.
19. Open the n8n human-review form.
20. Select the human decision.
21. Confirm the workflow completes.
22. Open SSMS and verify the latest Contract_Review_Audit record.

14. Troubleshooting

Gemini key error: Confirm GEMINI_API_KEY is set in the current PowerShell session.

Python service unavailable: Confirm contract_review_runner.py is running.

n8n cannot reach Python: Confirm the local Python review service is running.

SQL connection failure: Confirm SQL Server is running and the n8n Microsoft SQL credential is configured for AICLCRAgent.

Human-review form error: Open the form from the active n8n Wait node; the form URL is generated when the workflow reaches Wait.

15. Project Files

- `AI_Contract_Review_Agent.ipynb` — development, embeddings, retrieval and RAG workflow.
- `contract_review_runner.py` — Python contract review service.
- `app.py` — Streamlit application.
- `SQL/AICLCRAGENT.sql` — SQL Server scripts.
- `Standard Payment Terms Policy` — policy knowledge source.
- `Vendor Service Agreement` — sample contract knowledge source.
- `README.md` — project documentation.
- `.gitignore` — prevents local/secret files from being committed.

16. Security Notes

- Never commit Gemini API keys, SQL passwords, `.env` files or other secrets.
- Use environment variables for secrets.
- Use authorized reviewers for contract decisions.
- Use sanitized sample documents when sharing the project publicly.

17. Quick User Flow

START → Streamlit → n8n → Python/RAG → Gemini → AI Decision → Human Review → Workflow Update → SQL Server Audit